

The ENCODE Portal

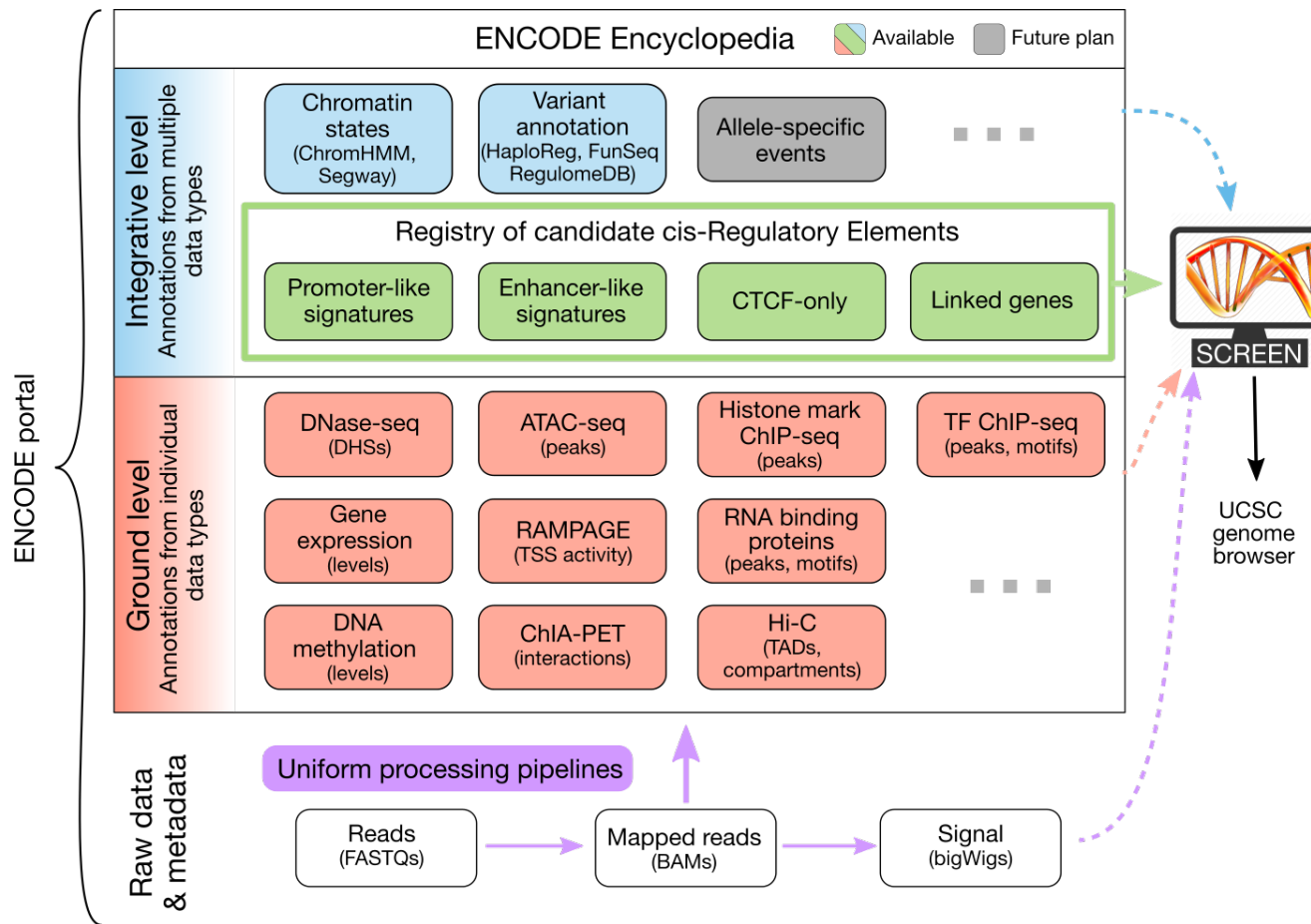
The ENCODE Portal

- The ENCODE Portal, developed and maintained by the Data Coordination Center (**ENCODE DCC**), is the canonical **source for all experimental metadata and data** from ENCODE and associated projects.
- The ENCODE Portal contains **raw and ground-level analysis data** generated by participating mapping centers using a wide-range of assays (integrative analysis data are available at the SCREEN portal).
- The Portal also **stores records of the materials and methods** used to perform the assays and subsequent analysis.
- **No account is needed** to view or download released data.

Portal quick links

- The ENCODE Portal contains the following types of data generated by the ENCODE Consortium:
 1. Data from a **wide-range of assays**, as well as protocol documents for each experiment
 2. **Antibody characterizations** that are performed as part of the antibody characterization process defined by the consortium
 3. **Reference filesets** used in the ENCODE uniform processing pipelines

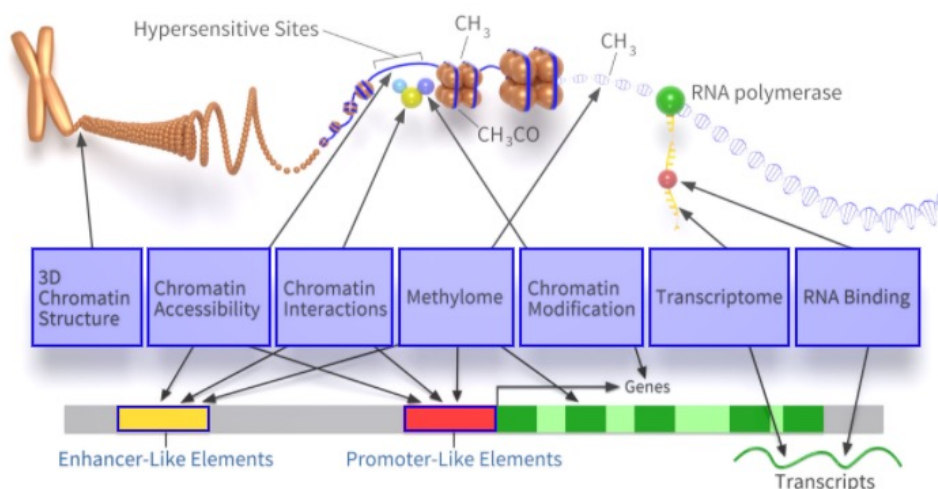
ENCODE Encyclopedia Version 5: Genomic and Transcriptomic Annotations



- The ENCODE Consortium not only **produces high-quality data**, but also **analyzes the data** in an integrative fashion. The ENCODE Encyclopedia organizes the most salient analysis products into annotations and **provides tools to search and visualize them**. The Encyclopedia has two levels of annotations:
- **Integrative-level annotations** integrate multiple types of experimental data and ground level annotations.
- **Ground-level annotations** are derived directly from the experimental data, typically produced by uniform processing pipelines.

ENCODE: Encyclopedia of DNA Elements

<https://www.encodeproject.org/>



Based on an image by Darryl Leja (NHGRI), Ian Dunham (EBI), Michael Pazin (NHGRI)

About ENCODE Project

Getting Started

Experiments

Search ENCODE portal ?

ENCODE Q

Functional Characterization Experiments

About ENCODE Encyclopedia

candidate Cis-Regulatory Elements

Search for candidate Cis-Regulatory Elements ?

Hosted by [SCREEN](#)

Human GRCh38 Q

Mouse mm10 Q

[Visit hg19 site](#)

HUMAN

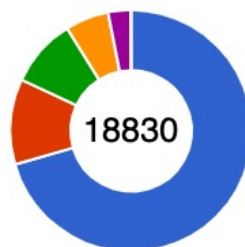
MOUSE

WORM

FLY

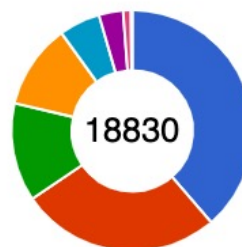
Data Matrix

Project



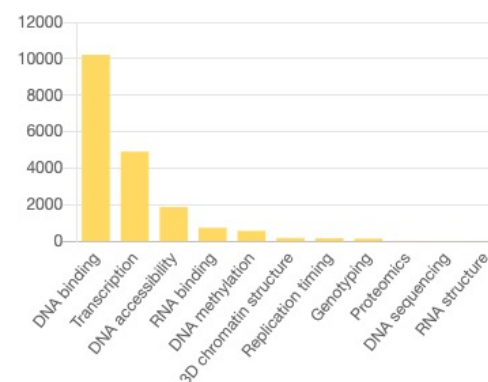
ENCODE
 Roadmap
 modERN
 modENCODE
 GGR
 community

Biosample Type



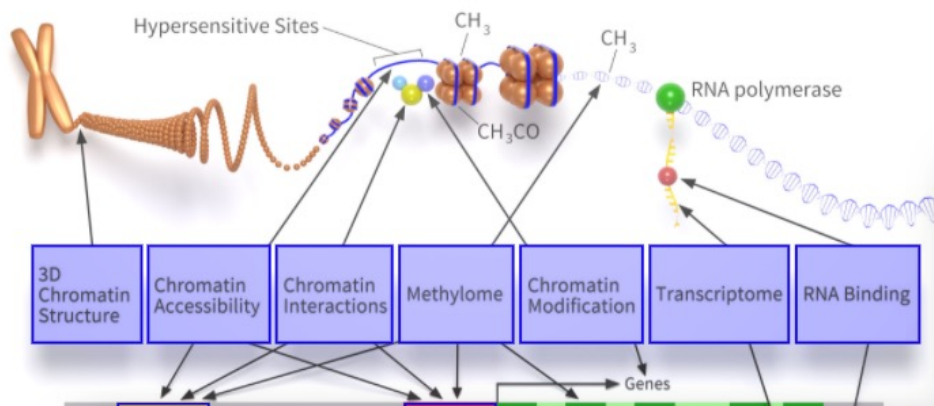
cell line
 tissue
 whole organisms
 primary cell
 single cell
 in vitro differentiated cells
 cell-free sample
 organoid

Assay Categories



? Help

ENCODE: Encyclopedia of DNA Elements

[About ENCODE Project](#)[Getting Started](#)[Experiments](#)

Search ENCODE portal ?

Getting Started

[Project overview](#) | [Using the portal](#) | [REST API](#) | [Data organization](#) | [Data submission](#) | [Encyclopedia](#)

Introduction to the Portal

Portal quick links

The ENCODE Portal contains the following types of data generated by the ENCODE Consortium:

- Data from a **wide-range of assays**, as well as protocol documents for each experiment
- **Antibody characterizations** that are performed as part of the **antibody characterization process** defined by the consortium
- **Reference filesets** used in the ENCODE uniform processing pipelines

Experiment search

Clear Filters

Assay type

DNA binding	10211
Transcription	4911
DNA accessibility	1873
RNA binding	737

Showing 25 of 18830 results

[Report](#) [Experiment matrix](#) [View All](#) [Download](#) [Visualize](#)

microRNA-seq of adrenal gland

Mus musculus C57BL/6JxCAST/EIJ adrenal gland tissue (4 days)

Lab: Ali Mortazavi, UCI

Project: ENCODE

Add all items

Experiment

ENC5R42

rele

Antibody lot search

Clear Filters

Eligibility status

awaiting characterization	1217
not pursued	1075
partially characterized	614
characterized to	

Target organism

Showing 25 of 3340 results

[Report](#) [View All](#)

goat IgG (isotype control)

Source: Sigma

Product ID / Lot ID: I9140 / 106K7431

Antibody

ENCAB330RYZ

released

H3K27me3 (*Homo sapiens*)

Source: Abcam

Product ID / Lot ID: A10096 / 1000000000

Antibody

ENCAB328EIA

released

Reference Sequences

Genome References

The ENCODE project uses Reference Genomes from [NCBI](#) or [UCSC](#) to provide a consistent framework for mapping high-throughput sequencing data. In general, ENCODE data are mapped consistently to 2 human (GRCh38, hg19) and 2 mouse (mm9/mm10) genomes for historical comparability. *Drosophila melanogaster* experiments are mapped to either dm3 or dm6 and *Caenorhabditis elegans* experiments are mapped to ce10 or ce11. The official reference files for each Uniform processing pipeline can be found in the table below, organized by organism and pipeline. In addition to the genome sequences (we generally use the "no alt" version for each genome), a variety of other crucial files can be found there as well (GENCODE transcript references, chromosome size files, the phage lambda genome, etc.).

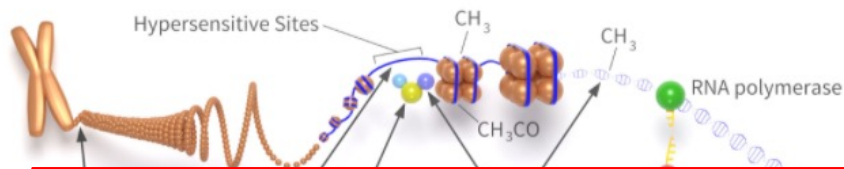
Antibody characterization process

Introduction

Many of the assays used in the ENCODE project involve the use of antibodies. Choosing an appropriate antibody involves a regimented and extensive antibody characterization process to establish a line of evidence that these antibodies specifically and accurately bind the target in question. It has long been a goal of the project to make this process and its results transparent.

[Basic organization](#) | [Data from previous and external projects](#)
[Antibody Lot Eligibility Statuses](#) | [Antibody Characterization Statuses](#) | [Antibody Standards](#)
[Search](#) | [How To](#)

ENCODE: Encyclopedia of DNA Elements

[About ENCODE Project](#)[Getting Started](#)[Experiments](#)

Search ENCODE portal ?

[ENCODE](#)[Functional Characterization Experiments](#)

Using the Portal

[Browse for data](#) | [Visualize data](#) | [Download files](#)

Browse and filter experiments

Clicking the “Search” option located in the “Data” toolbar menu located in the upper left corner brings up a list of all available experiments that have been used to generate ENCODE data. By default, search results are pre-filtered to experiments of status “released” (notice that in Figure 1, the facet term “released” is already highlighted). However, “archived” and “revoked” experiments are also publicly viewable. Explore the [status terminology page](#) for more information on what each status means.

The screenshot shows the ENCODE portal search results page. The top navigation bar includes links to ENCODE, Data, Encyclopedia, Materials & Methods, and Help. A search bar is located in the top right corner. The main content area displays search results for "scRNA-seq of A549". The left sidebar contains filters for Assay type, Assay title, Status, Perturbation, Target category, and Target of assay. The right sidebar contains a list of experiments with links to experiment pages, audits, and JSON format. Annotations highlight key features:

- Report, matrix, and summary views**: Points to the top of the search results section.
- Batch download**: Points to the "Download" button in the top right of the search results section.
- Search box**: Points to the search bar in the top right corner.
- Link to experiment page**: Points to the "Experiment" link in the right sidebar.
- View audits**: Points to the "Audits" link in the right sidebar.
- Add experiment to cart**: Points to the "Add to cart" button in the right sidebar.
- View in JSON format**: Points to the "JSON" link in the right sidebar.
- Facet sidebar**: Points to the left sidebar filters.

ENCODEDataEncyclopediaMaterials & MethodsHelp

Search...Q

Report, matrix, and summary views

Batch download

Search box

Experiment search

Showing 25 of 14938 results

View AllDownloadVisualize

Facet sidebar

Assay type

Assay title

TF ChIP-seq3640

Histone ChIP-seq3031

Control ChIP-seq2219

RNA-seq1172

ATAC-seq1071

DNase-seq769

Genome array252

Control eCLIP234

ecDNA225

Status

Selected filters: released

released14938

archived1077

revoked294

Perturbation

Selected filters: not perturbed

not perturbed14938

perturbed1938

Target category

Target of assay

scRNA-seq of A549

Homo sapiens A549

Lab: Michael Snyder, Stanford

Project: ENCODE

Control ChIP-seq of SK-N-SH

Homo sapiens SK-N-SH

Lab: Michael Snyder, Stanford

Project: ENCODE

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Control ChIP-seq of spleen

Homo sapiens spleen female adult (41 years)

Add all items to cart

View in JSON format

View audits

Add experiment to cart

Type a search term into the box

Select from multiple facets (logical AND)

Click items in "Selected filters" to undo filter

Target category

Target of assay

Selected filters: ✕ H3K4me3
✕ H3K4me2

Q H3K4

H3K4me3	206
H3K4me1	136
H3K4me2	51
H3K4ac	8

Organism

Selected filters: ✕ Homo sapiens

Homo sapiens	257
Mus musculus	18

Biosample classification

Selected filters: ✕ tissue

tissue	
cell line	120
primary cell	109
in vitro differentiated cells	28

Expand or collapse facets

Select multiple items in the same facet (logical OR)

Exclude items by clicking the red icon (logical NOT)

List view

ENCODE Data Encyclopedia Materials & Methods Help

Experiment search

Showing 25 of 14938 results

Assay type

Assay title

TF ChIP-seq 3640

Histone ChIP-seq 3031

Control ChIP-seq 2219

total RNA-seq 1172

scRNA-seq 1071

DNAse-seq 769

DNAse array 252

Control eCLIP 234

eCLIP 225

Status

Selected filters: released

released 14938

archived 1077

revoked 294

Perturbation

Selected filters: not perturbed

not perturbed 14938

perturbed 13938

Target category

Target of assay

H3K4me3 493

H3K4me1 418

H3K36me3 412

H3K27me3 393

Experiment

scRNA-seq of A549

Homo sapiens A549

Lab: Michael Snyder, Stanford

Project: ENCODE

Experiment

Control ChIP-seq of SK-N-SH

Homo sapiens SK-N-SH

Lab: Michael Snyder, Stanford

Project: ENCODE

Experiment

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Experiment

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Experiment

Control ChIP-seq of K562

Homo sapiens K562

Lab: Michael Snyder, Stanford

Project: ENCODE

Experiment

Control ChIP-seq of spleen

Homo sapiens spleen female adult (41 years)

Lab: Bradley Bernstein, Broad

Project: ENCODE

Experiment

Control ChIP-seq of heart left ventricle

Homo sapiens heart left ventricle female adult (66 years)

Lab: Michael Snyder, Stanford

Project: ENCODE

Summary view

ENCODE Data Encyclopedia Materials & Methods Help

Summary

Choose an organism:

Homo sapiens Mus musculus Drosophila melanogaster Caenorhabditis elegans

There are 11969 results.

Search list Tabular report Summary matrix

central nervous system circulatory system digestive system endocrine system excretory system

exocrine system immune system musculoskeletal system peripheral nervous system reproductive system

respiratory system sensory system skeletal system integumentary system

adrenal gland bone element brain bronchus colon esophagus gallbladder gonad intestine kidney limb liver mammary gland mouth muscle of body nose pancreas pericardium prostate gland skin of body spinal chord stomach testis thymus thyroid gland trachea

adipose tissue blood blood vessel bone marrow connective tissue embryo epithelium placenta lymphoid tissue lymph node lymphatic vessel

Lab

11969

Assay

11969

Status

Replication

unreplicated isogenic anisogenic

released archived revoked

Matrix view

ENCODE Data Encyclopedia Materials & Methods Help

Experiment Matrix

Filter the experiments included in the matrix:

Enter search term(s)

Showing 14938 results

ASSAY

cell line

K562 536 1700 541 246 14 444 122 2 2 149 67 162 104 3 116 7 17 101 17 1

HepG2 557 15 66 10 21 3 107 103 2 3 2 1 6 2 6 1 6 2 25

GM12878 185 15 24 10 2 2 3 2 6 1 1 7 2 6 1 6 2 2

HEK293 156 6 35 2 2 2 2 2 2 2 2 2 2 2 2 2 2

MCF-7 146 11 29 5 4 2 2 2 7 4 2 3 1 6 3

tissue

liver 42 91 29 23 8 1 7 1 9 2 7 7 1 1

stomach 17 72 26 20 21 3 4 4 10 5 6 4 1

heart 5 79 15 10 21 3 1 1 10 3 1 1 1

lung 6 58 15 12 16 2 4 1 7 1 4 8 1

forebrain 1 62 9 9 4 2 9 9 9 4 2 3 1 6 3

whole organisms

whole organism 996 987 109 19 3

carcass 996 987 109 19 3

primary cell

bone marrow-derived macrophage 66 499 122 247 8 149 38 16 24 7 16 65 15 37 30 1 12 27 7

keratinocyte 13 16 6 8 2 1 1 2 1 5 1 6

endothelial cell of umbilical vein 4 16 3 8 2 2 1 5 6

common myeloid progenitor, CD34-positive 11 9 1 12 1

B cell 2 25 8 4 6 1 1 1

Report view

ENCODE Data Encyclopedia Materials & Methods Help

Experiment report

Showing results 1 to 25 of 14938

Columns Download TSV

Assay type

Assay title

TF ChIP-seq 3640

Histone ChIP-seq 3031

Control ChIP-seq 2219

total RNA-seq 1172

scRNA-seq 1071

DNAse-seq 769

DNAse array 252

Control eCLIP 234

eCLIP 225

Status

Selected filters: released

released 14938

archived 1077

revoked 294

Perturbation

Selected filters: not perturbed

not perturbed 14938

perturbed 13938

Target category

Target of assay

H3K4me3 493

H3K4me1 418

H3K36me3 412

H3K27me3 393

ID	Accession	Assay name	Assay title	Target of assay	Target gene symbol	Biosample summary	Biosample term name	Description	Lab	Project	Status	Files	Related series	Biosample accession
ENC58514NTD	ENC58514NTD	single-cell RNA-seq	scRNA-seq of A549	A549	A549	Chip-seq on human A549	Michael Snyder, Stanford	ENC58514NTD	released	/files/ENC58514NTD/	ENC58514NTD			
ENC58514UWH	ENC58514UWH	Control ChIP-seq	Control ChIP-seq of SK-N-SH	SK-N-SH	SK-N-SH	Chip-seq on human SK-N-SH	Michael Snyder, Stanford	ENC58514UWH	released	/files/ENC58514UWH/	ENC58514UWH			
ENC58510PBT	ENC58510PBT	Control ChIP-seq	Control ChIP-seq of K562	K562	K562	Chip-seq on human K562	Michael Snyder, Stanford	ENC58510PBT	released	/files/ENC58510PBT/	ENC58510PBT			
ENC58502CWG	ENC58502CWG	Control ChIP-seq	Control ChIP-seq of K562	K562	K562	Chip-seq on human K562	Michael Snyder, Stanford	ENC58502CWG	released	/files/ENC58502CWG/	ENC58502CWG			
ENC585290GE	ENC585290GE	Control ChIP-seq	Control ChIP-seq of K562	K562	K562	Chip-seq on human K562	Michael Snyder, Stanford	ENC585290GE	released	/files/ENC585290GE/	ENC585290GE			
ENC58516LTS	ENC58516LTS	Control ChIP-seq	Control ChIP-seq of K562	K562	K562	Chip-seq on human K562	Michael Snyder, Stanford	ENC58516LTS	released	/files/ENC58516LTS/	ENC58516LTS			
ENC585149QP	ENC585149QP	Control ChIP-seq	Control ChIP-seq of spleen female adult (41 years)	spleen female adult (41 years)	spleen	Control ChIP-seq on spleen female adult (41 years)	Bradley Bernstein, Broad	ENC585149QP	released	/files/ENC585149QP/	ENC585149QP			
ENC585049PZ	ENC585049PZ	Control ChIP-seq	Control ChIP-seq of heart left ventricle female adult (66 years)	heart left ventricle female adult (66 years)	heart left ventricle	Control ChIP-seq on heart left ventricle female adult (66 years)	Bradley Bernstein, Broad	ENC585049PZ	released	/files/ENC585049PZ/	ENC585049PZ			

- [List view](#): displays results in a list. Each experiment is labeled with a summary of [assay](#) and [biosample](#) used.

Experiment search

Clear Filters ✕

Assay type

DNA binding	10211
Transcription	4911
DNA accessibility	1873
RNA binding	737

Assay title



TF ChIP-seq	4093
Histone ChIP-seq	3311
Control ChIP-seq	2447
DNase-seq	1418
scRNA-seq	1065
total RNA-seq	790
polyA plus RNA-seq	782
shRNA RNA-seq	658
ATAC-seq	362

Status

Selected filters: ✕ released

☒ released	18830
☐ archived	1103
☐ revoked	362

Showing 25 of 18830 results

Report

Experiment matrix

View All

Download

Visualize

{ : }

Add all items to cart

microRNA-seq of adrenal gland

Mus musculus C57BL/6JxCAST/EiJ adrenal gland tissue (4 days)

Lab: Ali Mortazavi, UCI

Project: ENCODE

Experiment



ENCSR424VMF

● released

ChIA-PET of A549

Homo sapiens A549

Target: POLR2A

Lab: Yijun Ruan, JAX

Project: ENCODE

Experiment



ENCSR138NSW

● released

● 2

DNase-seq of head of caudate nucleus

Homo sapiens head of caudate nucleus tissue female adult (90 or above years)

Lab: John Stamatoyannopoulos, UW

Project: ENCODE

Experiment



ENCSR688AWP

● released

DNase-seq of head of caudate nucleus

Homo sapiens head of caudate nucleus tissue male adult (86 years)

Lab: John Stamatoyannopoulos, UW

Project: ENCODE

Experiment



ENCSR860UTH

● released

DNase-seq of head of caudate nucleus

Homo sapiens head of caudate nucleus tissue female adult (90 or above years) with Cognitive impairment

Lab: John Stamatoyannopoulos, UW

Project: ENCODE

Experiment



ENCSR516RTV

● released

- Report view: displays results in tabular format, with a default selection of metadata properties as the columns.

Experiment report

[Clear Filters](#)

Assay type

DNA binding	10211
Transcription	4911
DNA accessibility	1873
RNA binding	737

Assay title



TF ChIP-seq	4093
Histone ChIP-seq	3311
Control ChIP-seq	2447
DNase-seq	1418
scRNA-seq	1065
total RNA-seq	790
polyA plus RNA-seq	782
shRNA RNA-seq	658
ATAC-seq	362



List



Experiment matrix



Columns

Download TSV

Items per page:

25 50 100 200

< 1 2 3 4 5 6 7 ... 754 >

Showing results 1 to 25 of 18830

ID	Accession	Assay name	Assay title	Target of assay	Target gene symbol	Biosample summary	Biosam
ENCSR424VMF	ENCSR424VMF	microRNA-seq	microRNA-seq			C57BL/6JxCAST/EiJ adrenal gland tissue (4 days)	adrenal
ENCSR138NSW	ENCSR138NSW	ChIA-PET	ChIA-PET	POLR2A	POLR2A	A549	A549

- Matrix view: displays results in a matrix, organized with **biosamples** along the y-axis and assay type along the x-axis.

Experiment Matrix



Clear Filters

Filter the experiments included in the matrix:

Showing 18830 results

Enter search term(s)



Assay type

DNA binding	10211
Transcription	4911
DNA accessibility	1873
RNA binding	737

Assay title

Search

TF ChIP-seq	4093
Histone ChIP-seq	3311
Control ChIP-seq	2447
DNase-seq	1418
scRNA-seq	1065
total RNA-seq	790
polyA plus RNA-seq	782
shRNA RNA-seq	658
ATAC-seq	362

Status

Selected filters: released

released	18830
archived	1103
revoked	362

ASSAY →

← BIOSAMPLE

	TF ChIP-seq	Histone ChIP-seq	Control ChIP-seq	DNase-seq	scRNA-seq	total RNA-seq	polyA plus RNA-seq	shRNA RNA-seq	ATAC-seq	Mint-ChIP-seq	microRNA-seq	DNAme array	Control eCLIP
cell line	2655	750	702	209	2	114	185	553	123	16	26	91	232
K562	634	19	186	7		13	19	285	2		2	3	125
HepG2	652	15	76	2		6	11	268	2		2	3	107
A549	243	86	77	14			27		6			2	
GM12878	188	15	27	2	2	5	13		2		2	3	
MCF-7	148	18	34	7		1	4		1		2	2	
tissue	332	1793	567	703	14	205	397		181		180	122	2
liver	42	91	29	15		3	20		7		7	1	
stomach	17	72	26	22		5	15		6		4	3	
heart	5	79	15	21		3	16		7		9		
spleen	18	58	23	6		8	11		1		3	4	
lung	6	58	15	17		2	11		5		4	2	
whole organisms	996		987			272	80	105					
whole organism	996		987			268	72	105					
carcass						4	8						

- Summary view: displays a general overview of the data in chart form. A body map diagram is also available for human data.

Summary

 Homo sapiens

 Mus musculus

 Drosophila melanogaster

 Caenorhabditis elegans

There are 12825 results.

[Clear Filters](#)

[List](#)

[Report](#)

[Experiment matrix](#)

central nervous system

exocrine system

respiratory system

circulatory system

immune system

sensory system

digestive system

musculature

skeletal system

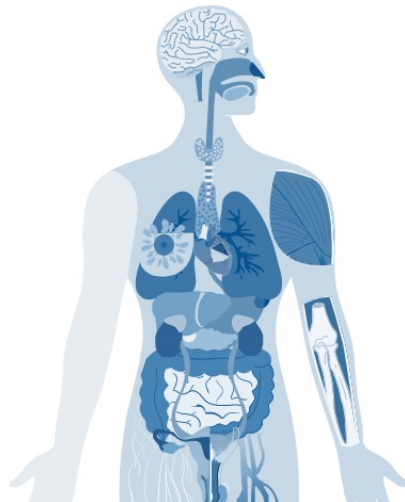
endocrine system

peripheral nervous system

integumental system

excretory system

reproductive system



adrenal gland

bone element

breast

colon

eye

gonad

intestine

large intestine

liver

mammary gland

musculature of body

nose

pancreas

pericardium

skeleton

small intestine

spleen

arterial blood vessel

brain

bronchus

esophagus

gallbladder

heart

kidney

limb

lung

mouth

nerve

ovary

penis

prostate gland

skin of body

spinal chord

stomach



[Clear body map selections](#)

adipose tissue

blood

blood vessel

bone marrow

connective tissue

embryo

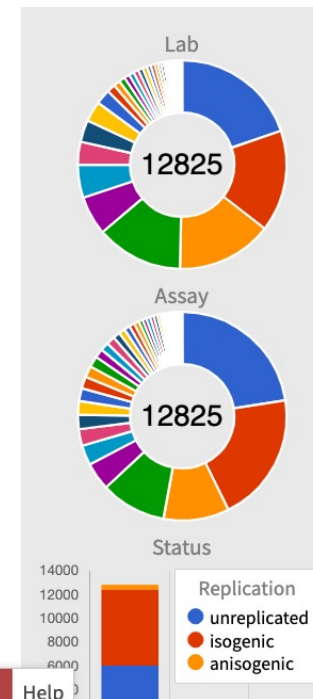
epithelium

placenta

lymphoid tissue

lymph node

lymphatic vessel



1. Data from a wide-range of assays

- <https://www.encodeproject.org/search/?type=Experiment&status=released>

Experiment search

Clear Filters ✕

Showing 25 of 18830 results

Report Experiment matrix View All Download Visualize

{:}

Assay type

Assay title

Search

TF ChIP-seq	4093
Histone ChIP-seq	3311
Control ChIP-seq	2447
DNase-seq	1418
scRNA-seq	1065
total RNA-seq	790
polyA plus RNA-seq	782
shRNA RNA-seq	658
ATAC-seq	362

Status

Selected filters

Add all items to cart

microRNA-seq of adrenal gland

Mus musculus C57BL/6JxCAST/EiJ adrenal gland tissue (4 days)

Lab: Ali Mortazavi, UCI

Project: ENCODE

Experiment

ENCSR424VMF

● released

ChIA-PET of A549

Homo sapiens A549

Target: POLR2A

Lab: Yijun Ruan, JAX

Project: ENCODE

Experiment

ENCSR138NSW

● released

2

DNase-seq of head of caudate nucleus

Homo sapiens head of caudate nucleus tissue female adult (90 or above years)

Lab: John Stamatoyannopoulos, UW

Project: ENCODE

Experiment

ENCSR688AWP

● released

1. Data from a wide-range of assays

Experiments / [ChIA-PET](#) / [Homo sapiens](#) / [A549](#)



Experiment summary for ENCSR138NSW

2

{;}

Summary		Attribution			ENCODE PHASE 4
Status:	 released	Lab:	Yijun Ruan, JAX		
Assay:	ChIA-PET	Award:	UM1HG009409 (Yijun Ruan, JAX)		
Target:	POLR2A	Project:	ENCODE		
Biosample summary:	<i>Homo sapiens</i> A549	Aliases:	yijun-ruan:A549_RNAPII		
Biosample Type:	cell line	Date submitted:	January 12, 2021		
Replication type:	isogenic	Date released:	January 12, 2021		
Description:	RNAPII ChIA-PET in A549				
Nucleic acid type:	DNA				
Size range:	350-600				
Extraction method:	SDS				
Size selection method:	BluePippin				
Platform:	Illumina NovaSeq 6000				
Controls:	ENCSR414IDX				

1. Data from a wide-range of assays

Isogenic replicates					
Isogenic replicate	Technical replicate	Summary	Biosample	Antibody	Library
1	1	A549 cell line	ENCBS085MQM	ENCAB725WBH	ENCLB877LSQ
2	1	A549 cell line	ENCBS744PGM	ENCAB725WBH	ENCLB430DDY

ENCBS085MQM / cell line

Summary	Attribution
Status: released	Lab: Yijun Ruan, JAX
Term name: A549	Award PI: Yijun Ruan, JAX
Term ID: EFO:0001086	Submitted by: Daniel Capurso
Summary: <i>Homo sapiens</i> A549 cell line	Source: ATCC
Description: Human epithelial cell line derived from a lung carcinoma tissue. The cells are adherent in culture. The karyotype is hypotriploid male with a modal chromosome number of 12. There are numerous chromosome abnormalities and marker chromosomes.	Project: ENCODE
	External resources: Cellosaurus:CVCL_0023
	Aliases: yijun-ruan:A549_cell_culture_biorep1

Donor information

Status: released
Accession: ENCD0000AAZ
Aliases: encode:donor of A549, bradley-bernstein:Donor of A549 cells, john-stamatoyannopoulos:ATCC_A549
Species: <i>Homo sapiens</i>
Life stage: Adult
Age: 58 years
Sex: Male
Ethnicity: Caucasian
External resources: GEO:SAMN05733878
References: PMID:4357758

Functional genomics experiments using this biosample

Accession	Assay	Biosample term name	Target	Description	Lab
ENCSR138NSW	ChIA-PET	A549	POLR2A	RNAPII ChIA-PET in A549	Yijun Ruan, JAX
ENCSR911ZMB	ChIA-PET	A549	CTCF	CTCF ChIA-PET in A549	Yijun Ruan, JAX

Displaying 2 of 2

1. Data from a wide-range of assays

Files

Genome browser | Association graph | **File details**

Include deprecated files ☐

Choose analysis
Lab custom GRCh38 ▾

Filter files

File format

fastq	4
bam	4
hic	2
bigWig	2
bed broadPeak	2
bigBed broadPeak	2
bigInteract	2
bedpe	2
bed bedGraph	2

Output type

reads	4
signal of unique reads	4
peaks	4
long range chromatin interactions	4
chromatin interactions	2
alignments	2
unfiltered alignments	2

Replicates

1	11
2	11

Replicate 1

ENCFF062QHI (reads) → filtering classification alignment → ENCF173LRG (unfiltered alignments) → chromatin loop identification → ENCF346FGU (long range chromatin interactions) → file format conversion → ENCF429CCB (long range chromatin interactions)

ENCFF815UQU (reads) → filtering classification alignment → ENCF225RTS (alignments) → signal generation → ENCF652UBC (signal of unique reads) → file format conversion → ENCF336JPS (signal of unique reads)

ENCFF196FLV → filtering classification alignment → ENCF225RTS (alignments) → file format conversion interaction calling → ENCF619KTL (chromatin interactions)

ENCFF244WBE → filtering classification alignment → ENCF225RTS (alignments) → peak calling → ENCF244JPS (peaks) → file format conversion → ENCF735NOT (peaks)

ENCFF66KAN → filtering classification alignment → ENCF225RTS (alignments) → peak calling → ENCF244JPS (peaks) → file format conversion → ENCF735NOT (peaks)

Replicate 2

ENCFF126WVE (reads) → filtering classification alignment → ENCF724GKE (unfiltered alignments) → signal generation → ENCF746MHF (signal of unique reads) → file format conversion → ENCF737FHQ (signal of unique reads)

ENCFF634ZUN (reads) → filtering classification alignment → ENCF724GKE (unfiltered alignments) → peak calling → ENCF633NGQ (peaks) → file format conversion → ENCF478ZK (peaks)

ENCFF754BBD (alignments) → chromatin loop identification → ENCF421KXP (long range chromatin interactions) → file format conversion → ENCF251MPG (long range chromatin interactions)

ENCFF634ZUN (reads) → filtering classification alignment → ENCF724GKE (unfiltered alignments) → file format conversion interaction calling → ENCF421RLB (chromatin interactions)

Download Graph

1. Data from a wide-range of assays

Documents

Pipeline Protocol

Description: Ruan Lab ChIA-PIPE processing pipeline for ChIA-PET data



 [ENCODE_ChIA_PET_pipeline_readme_12172021](#) 

General Protocol

Description: Ruan Lab in-situ ChIA-PET protocol



 [In-situ chiapet_Ruan Lab_final.pdf](#) 

ENCODE, ChIA-PET Data processing pipeline description for Ruan Lab

ChIA-PIPE is a fully automated data processing pipeline to process ChIA-PET data for both in-situ ChIA-PET and long-read ChIA-PET (Lee et al., 2020 Genome Biology. doi:10.1126/sciadv.aay2078). The complete code is available in github site (<https://github.com/TheJacksonLaboratory/ChIA-PIPE>).

1. Pipeline install and launch the process

After users download and install ChIA-PIPE from the github site, they can fill in pipeline parameters in a config file, and then launch ChIA-PIPE on HPC. Key pipeline parameters in a config file follow, using a GM10248 CTCF ChIA-PET experiment as an example (ENCODE accession: ENCSCR627XQX):

```
# The name of the sequencing run
run="LHG0047H"
```

```
# The type of sequencing run:
# "miseq" - around 30 million reads
# "hiseq" - around 300 million reads
# "pooled" - around 1 billion reads
# It adjusts a proper computing resource (processors, memory, and walltime).
run_type="hiseq"
```

In situ ChIA-PET protocol (Ruan Lab 18/06/2020)

Background

ChIA-PET is a proven method to map protein-mediated chromatin interactions. The initial ChIA-PET method was developed based ChIP-PET [Wei et al., 2006] for mapping transcription factor (TF) genome-wide occupancy by adding a proximity ligation to the ChIPed chromatin material, so to establish the connectivity between chromatin fragments tethered together by protein factors [Fullwood et al., 2009]. The original ChIA-PET protocol employed dual linkers in proximity ligation to distinguish non-specific ligation and a type IIS restriction enzyme digestion (MmeI) to generate a 20bp tag of chromatin DNA on each side of the dual-linker ligation products, and the paired-end-tag (PET, tag-linker-tag) templates were subjected for high throughput paired end sequencing and mapping to demarcated long-range chromatin interactions [Fullwood et al., 2010]. To increase the mapping PET mapping efficiency, we employed random digestion by Tn5 transposase to the proximity ligation products to generate longer templates (tagmentation) for 2X250 bp PET sequencing [Li et al., 2017]. We have shown that long-read ChIA-PET significantly increased mapping efficiency and allow for detection of allele-specific chromatin interactions at single-base-pair resolution [Tang et al., 2015]. Inspired by in situ Hi-C protocol, we have now incorporated the steps of in situ digestion and in situ proximity ligation into the ChIA-PET protocol, so the in situ ChIA-PET, which showed significant reduction of non-specific proximity ligation and reduced the required cell numbers as low as 1 million cells per ChIA-PET experiment comparing to the previous ChIA-PET protocol. We have applied in situ ChIA-PET protocol in the ENCODE project and have to generated more than hundred ChIA-PET datasets associated with CTCF and RNAPII from ENCODE biosamples. Here, we describe the outline and the detailed step-by-step procedure of the in situ ChIA-PET protocol.

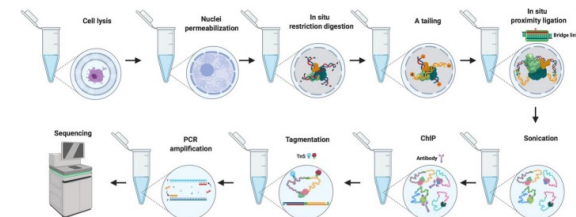


Figure 1. Schematic of in situ ChIA-PET.

2. Antibody characterizations

- <https://www.encodeproject.org/search/?type=AntibodyLot&status=released>

Antibody lot search

Clear Filters ✕

Showing 25 of 3340 results

Report

View All

{:}

Eligibility status

☒ awaiting characterization	1217
☐ not pursued	1075
☒ partially characterized	614
☐ characterized to	

Target organism

<i>Homo sapiens</i>	2910
<i>Drosophila melanogaster</i>	229
<i>Caenorhabditis elegans</i>	140
<i>Mus musculus</i>	121

Target category

Target of antibody

Search

H3K4me3	46
H3K27me3	38
H3K36me3	29
H3K9me3	29
H3K4me1	28
H3K27ac	27
CTCF	21
H3K4me2	16
EP300	11

goat IgG (isotype control)

Source: Sigma
Product ID / Lot ID: I9140 / 106K7431

Antibody
ENCAB330RYZ
● released

H3K27me3 (*Homo sapiens*)

Source: Abcam
Product ID / Lot ID: ab192985 / GR223657

Antibody
ENCAB328EIA
● released

H3K27me3 (*Homo sapiens*)

Source: Cell Signaling
Product ID / Lot ID: 9733 / 14

Antibody
ENCAB009VWX
● released

H3K9me3 (*Homo sapiens*)

Source: Abcam
Product ID / Lot ID: ab176916 / GR3218257

Antibody
ENCAB003LHL
● released

H3K9me3 (*Homo sapiens*)

Source: Abcam
Product ID / Lot ID: ab8898 / GR3244172-1

Antibody
ENCAB679IZV
● released

SMC3 (*Homo sapiens*)

Source: Abcam
Product ID / Lot ID: ab9263 / GR290533-7

Antibody
ENCAB501DLL
● released

RAD21 (*Homo sapiens*)

Source: Abcam
Product ID / Lot ID: ab992 / GR214359-8

Antibody
ENCAB695QDA
● released

2. Antibody characterizations

- <https://www.encodeproject.org/search/?type=AntibodyLot&status=released>

Antibodies / *Homo sapiens* / HIST2H3A + HIST2H3D + HIST2H3C

ENCAB328EIA

Antibody against *Homo sapiens* H3K27me3

{:}

Homo sapiens

any cell type or tissue

Ⓢ characterized to standards with exemption

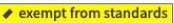
Status:	☒ released
Source (vendor):	Abcam
Product ID:	ab192985
Lot ID:	GR223657
Characterized targets:	H3K27me3 (<i>Homo sapiens</i>)
Host:	Rabbit
Clonality:	Monoclonal
Aliases:	bradley-bernstein:PchAb 1393

Functional genomics experiments using this antibody

Accession	Assay	Biosample term name	Target	Description	Lab
ENCSR024JJL	ChIP-seq	esophagus squamous epithelium	H3K27me3	H3K27me3 ChIP-seq on esophagus squamous epithelium male adult (54 years)	Bradley Bernstein, Broad

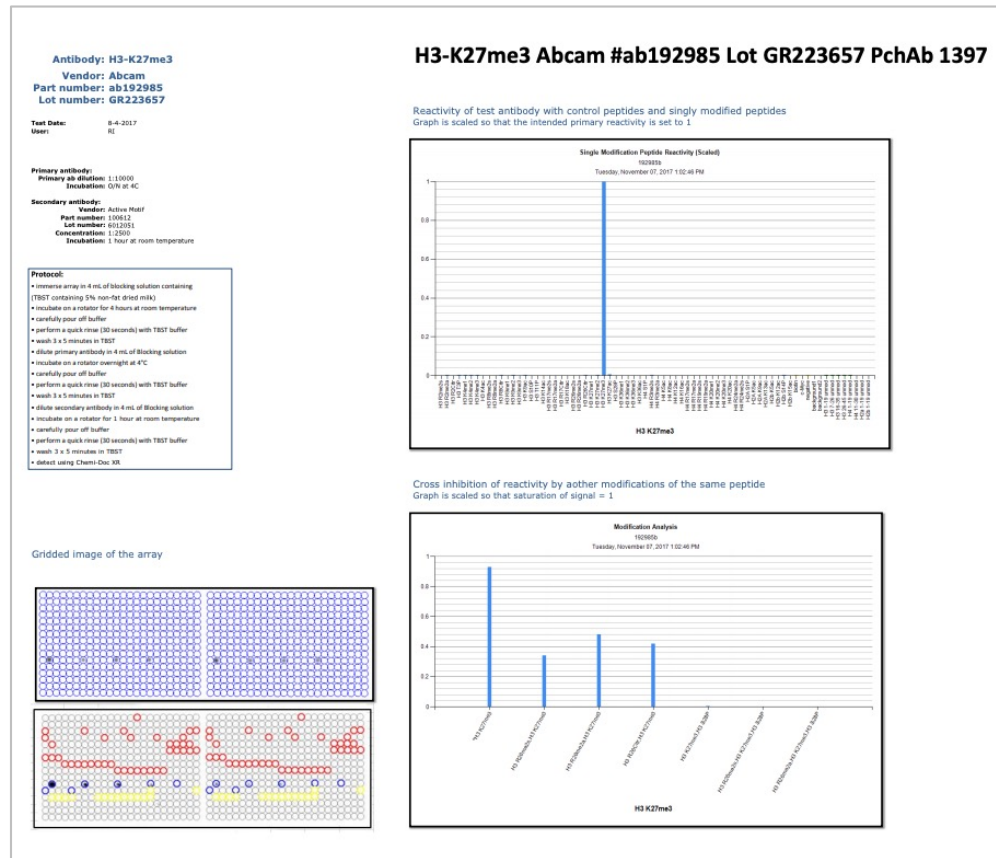
Displaying 1 of 1

Characterizations

H3K27me3 (<i>Homo Sapiens</i>) Method: dot blot assay   +	H3K27me3 (<i>Homo Sapiens</i>) esophagus squamous epithelium Method: immunoblot   +
---	--

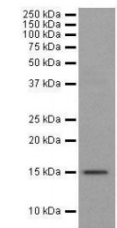
2. Antibody characterizations

- <https://www.encodeproject.org/search/?type=AntibodyLot&status=released>



Antibody: H3K27me3
Vendor: Abcam
Immunogen: Synthetic peptide within Human Histone H3 aa 1-100 (tri methyl K27). The exact sequence is proprietary. Database link: P68431
Product Number: ab192985
Lot Number: GR223657

Western Blot



Anti-Histone H3 (tri methyl K27) antibody [EPR18607] - ChIP Grade (ab192985) at 1/1000 dilution + NIH/3T3 (Mouse embryonic fibroblast cell line) whole cell lysate at 10 µg

Secondary
 Goat Anti-Rabbit IgG H&L (HRP) (ab97051) at 1/50000 dilution

Predicted band size: 15 kDa

Observed band size: 15 kDa

Exposure time: 1 second

Blocking/Dilution buffer: 5% BSA/TBST

Supplied by Abcam

3. Reference filesets

- <https://www.encodeproject.org/data-standards/reference-sequences/>

Reference Sequences

Genome References

The ENCODE project uses Reference Genomes from [NCBI](#) or [UCSC](#) to provide a consistent framework for mapping high-throughput sequencing data. In general, ENCODE data are mapped consistently to 2 human (GRCH38, hg19) and 2 mouse (mm9/mm10) genomes for historical comparability. *Drosophila melanogaster* experiments are mapped to either dm3 or dm6 and *Caenorhabditis elegans* experiments are mapped to ce10 or ce11. The official reference files for each Uniform processing pipeline can be found in the table below, organized by organism and pipeline. In addition to the genome sequences (we generally use the "no alt" version for each genome), a variety of other crucial files can be found there as well (GENCODE transcript references, chromosome size files, the phage lambda genome, etc.).

Reference File Sets

The table below includes files used by each pipeline for uniform processing by the ENCODE DCC, with associated details on genome assembly and annotation, if applicable. For your convenience, the GRC genome assembly and GENCODE annotation files are directly linked below. For further information, please contact encode-help@lists.stanford.edu

ENCODE4 Uniform Processing Pipeline Filesets

Organism	Pipeline(s)	Reference file set	Genome assembly & annotation
	ATAC-seq	ENCSR938RZZ	GRCh38 with GENCODE version V29
	Bulk RNA-seq	ENCSR151GDH	GRCh38 with GENCODE version V29
	ChIP-seq	ENCSR174MOP (TF, Histone)	GRCh38
		ENCSR487FYI (MINT-ChIP)	GRCh38
	DNase-seq	ENCSR083JOX	GRCh38
	Long read RNA-seq	ENCSR925QOG	GRCh38 with GENCODE version V29
	microRNA-seq	ENCSR608ULQ	GRCh38 with GENCODE version V29

Pipeline search

Showing 4 of 4 results

Clear Filters

Report

Assay names

Status

Title

Software

Lab

accession

Selected filters: [ENCL612HIG](#) [ENCL809GEM](#) [ENCL367MAS](#) [ENCL481MLO](#)

[ENCL612HIG](#)

[ENCL809GEM](#)

[ENCL367MAS](#)

[ENCL481MLO](#)

Histone ChIP-seq 2 (unreplicated)

Pipeline ENCL809GEM

Assays: ChIP-seq, Mint-ChIP-seq

Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, overlap_peaks.py, bedToBigBed

[released](#)

Histone ChIP-seq 2

Pipeline ENCL612HIG

Assays: ChIP-seq, Mint-ChIP-seq

Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, overlap_peaks.py, bedToBigBed

[released](#)

Transcription factor ChIP-seq 2

Pipeline ENCL367MAS

Assays: ChIP-seq, Mint-ChIP-seq

Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, Irreproducible Discovery Rate (IDR), Phantompeakqualtools, bedToBigBed

[released](#)

Transcription factor ChIP-seq 2 (unreplicated)

Pipeline ENCL481MLO

Assays: ChIP-seq, Mint-ChIP-seq

Software: Bowtie 2, Samtools, Sambamba, Picard, cutadapt, ptools, MACS, BEDTools, Irreproducible Discovery Rate (IDR), Phantompeakqualtools, bedToBigBed

[released](#)

3. Reference filesets

- <https://www.encodeproject.org/data-standards/reference-sequences/>

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		ENCSR487FYI (MINT-ChIP)	GRCh38
	DNase-seq	ENCSR083JOX	GRCh38
	Long read RNA-seq	ENCSR925QOG	GRCh38 with GENCODE version V29
	microRNA-seq	ENCSR608ULQ	GRCh38 with GENCODE version V29

Summary for reference file set ENCSR938RZZ

</

Training

- Factor: CTCF, RNAPII, p53
- Cell line: A549
- Data:
 - ChIP-seq
 - RNA-seq
 - ChIA-PET
 - HI-C
 - ATAC-seq

